

Fake SSH Honeyypot Attack Detection & Monitoring

Monitoring & Logging
Assignment

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Date: 10 April 2025

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FAKE SSH SERVER ATTACK



Objective: *Simulate an external SSH brute-force attack using a fake SSH server (Cowrie) and monitor it using centralized logging tools such as logwatch, ntopng, librenms.*

Cowrie is an open-source **honeypot** designed to simulate an SSH and Telnet server, primarily used for capturing and logging malicious activity in an environment. It emulates vulnerable services to attract attackers, logging their commands and actions, allowing security professionals to analyze attack patterns and techniques.

Details:

- A fake SSH honeypot (Cowrie) was deployed on ``nmshost``.
- ``webhost`` simulated the attacker using brute-force (Hydra).
- Logging and monitoring tools were configured to detect and visualize the attack.
- Such as logwatch , ntopng,etc.

01

Attack Description

Let's have a look



In this attack, a fake SSH server (such as cowrie or a custom netcat-based listener) is deployed on a system to simulate an open SSH port.

The goal is to trap unauthorized access attempts, gather attacker credentials, and analyze their behavior.

This form of attack is also known as a honeypot strategy. It listens on port 2222 or a different port (like commonly used 22 for realism). Logs connection attempts, usernames, and sometimes passwords. Helps identify brute-force attempts or attacker intentions.

02

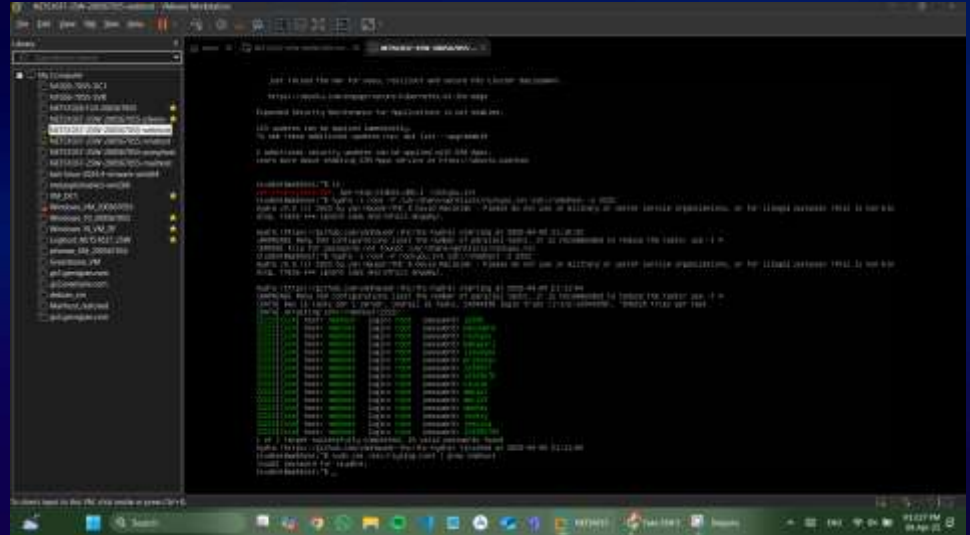
Attack performed

SSH from webhost

Doing the Brute Force Attack:

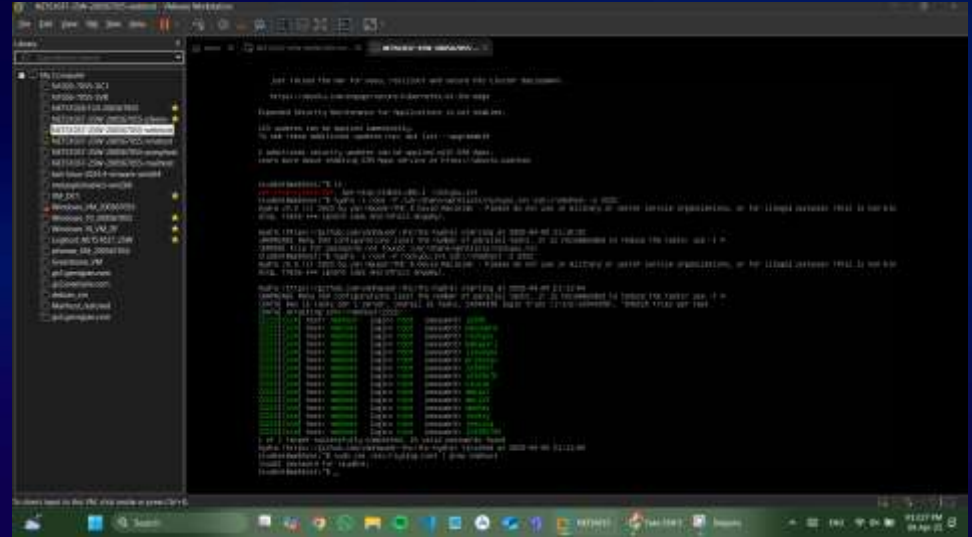
I tried to do the brute force attack(SSH) one from webhost with the help of hydra:

Command: **hydra -l root -P rockyou.txt ssh://nmshost -s 2222**

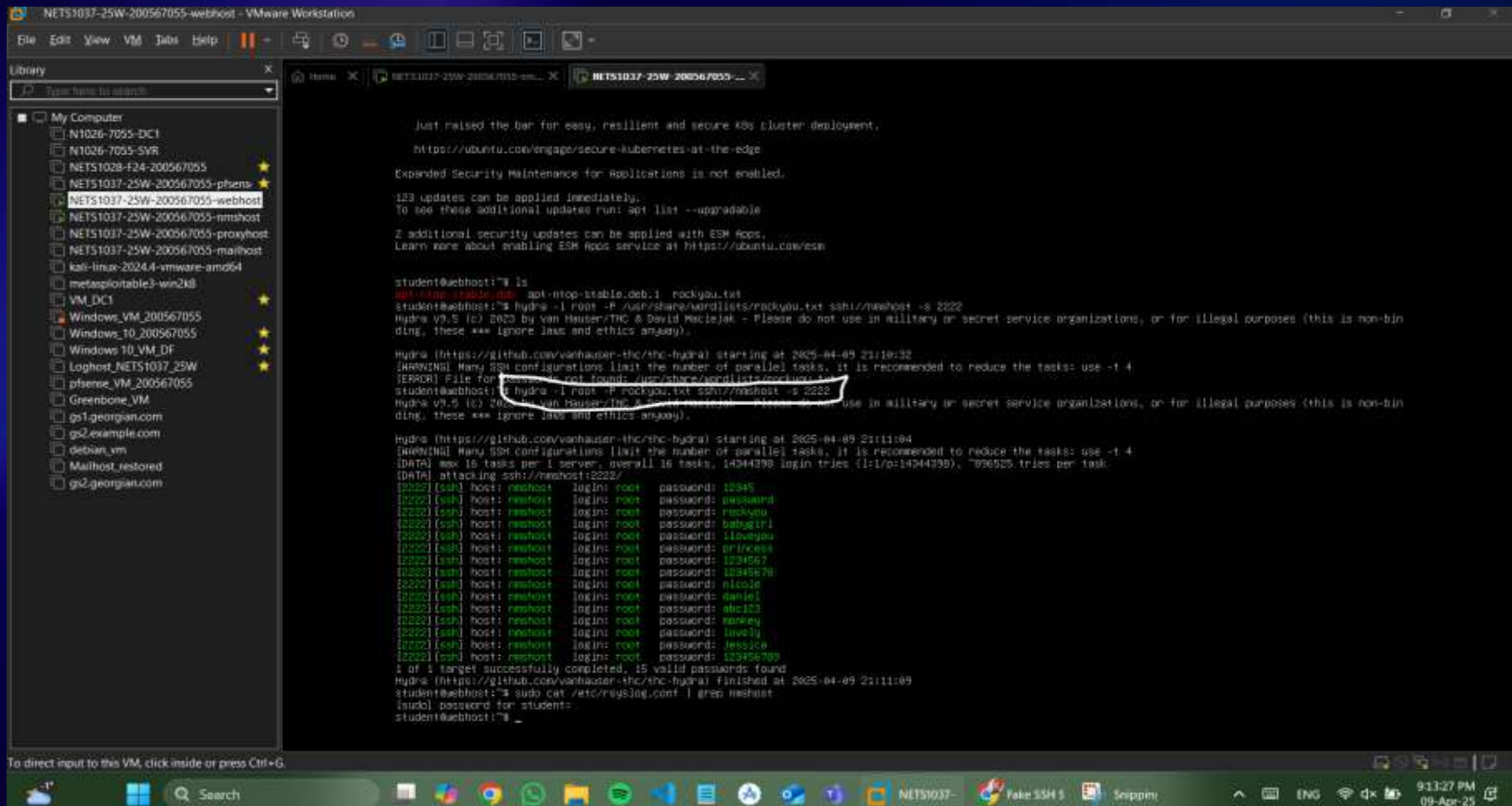


SSH from webhost

SSH did was simulating a brute-force or login attempts from another VM (like webhost) using hydra (SSH attempts) .



SSH from webhost using Hydra:



NETS1037-25W-200567055-webhost - VMware Workstation

File Edit View VM Tabs Help

Library

My Computer

- N1026-7055-DC1
- N1026-7055-SVR
- NETS1028-F24-200567055
- NETS1037-25W-200567055-phense
- NETS1037-25W-200567055-webhost
- NETS1037-25W-200567055-nmshost
- NETS1037-25W-200567055-proxyhost
- NETS1037-25W-200567055-mailhost
- kali-linux-2024.4-vmware-amd64
- metasploitable3-win2k8
- VM_DC1
- Windows_VM_200567055
- Windows_10_200567055
- Windows_10_VM_DF
- Loghost_NETS1037_25W
- phense_VM_200567055
- Greenbone_VM
- gs1.georgian.com
- gs2.example.com
- debian_vm
- Mailhost_restored
- gs2.georgian.com

Just raised the bar for easy, resilient and secure K8s cluster deployment.

<https://ubuntu.com/engage/secure-kubernetes-at-the-edge>

Expanded Security Maintenance for Applications is not enabled.

123 updates can be applied immediately.

To see these additional updates run: `apt list --upgradable`

2 additional security updates can be applied with ESM Apps.

Learn more about enabling ESM Apps service at <https://ubuntu.com/esm>

```
student@webhost:~$ ls
apt-get-stable.deb  apt-ntop-stable.deb.1  rockyou.txt
student@webhost:~$ hydra -i root -P /usr/share/wordlists/rockyou.txt ssh://nmshost -s 2222
Hydra V9.5 (c) 2023 by Van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is non-bin
ding, these *** ignore laws and ethics anyway!).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2025-04-09 21:10:32
[WARNING] Many SSH configurations limit the number of parallel tasks, it is recommended to reduce the tasks: use -i 4
[ERROR] File for password not found: /usr/share/wordlists/rockyou.txt
student@webhost:~$ hydra -i root -P rockyou.txt ssh://nmshost -s 2222
Hydra V9.5 (c) 2023 by Van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is non-bin
ding, these *** ignore laws and ethics anyway!).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2025-04-09 21:11:04
[WARNING] Many SSH configurations limit the number of parallel tasks, it is recommended to reduce the tasks: use -i 4
[DATA] max 16 tasks per 1 server, overall 16 tasks, 14044398 login tries (1:1/p:14044398), 796525 tries per task.
[DATA] attacking ssh://nmshost:2222/
[2222] ssh host: nmshost login: root password: 12345
[2222] ssh host: nmshost login: root password: password
[2222] ssh host: nmshost login: root password: rockyou
[2222] ssh host: nmshost login: root password: b4f0g1r1
[2222] ssh host: nmshost login: root password: 1loveyou
[2222] ssh host: nmshost login: root password: princeps
[2222] ssh host: nmshost login: root password: 1234567
[2222] ssh host: nmshost login: root password: 12345678
[2222] ssh host: nmshost login: root password: nicole
[2222] ssh host: nmshost login: root password: daniel
[2222] ssh host: nmshost login: root password: abc123
[2222] ssh host: nmshost login: root password: monkey
[2222] ssh host: nmshost login: root password: love1
[2222] ssh host: nmshost login: root password: jessica
[2222] ssh host: nmshost login: root password: 123456789
1 of 1 target successfully completed, 15 valid passwords found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2025-04-09 21:11:09
student@webhost:~$ sudo cat /etc/passwd | grep nmshost
(sudo) password for student:
student@webhost:~$
```

To direct input to this VM, click inside or press Ctrl+G.

NETS1037- Fake SSH S Snipping

9:13:27 PM 09-Apr-25

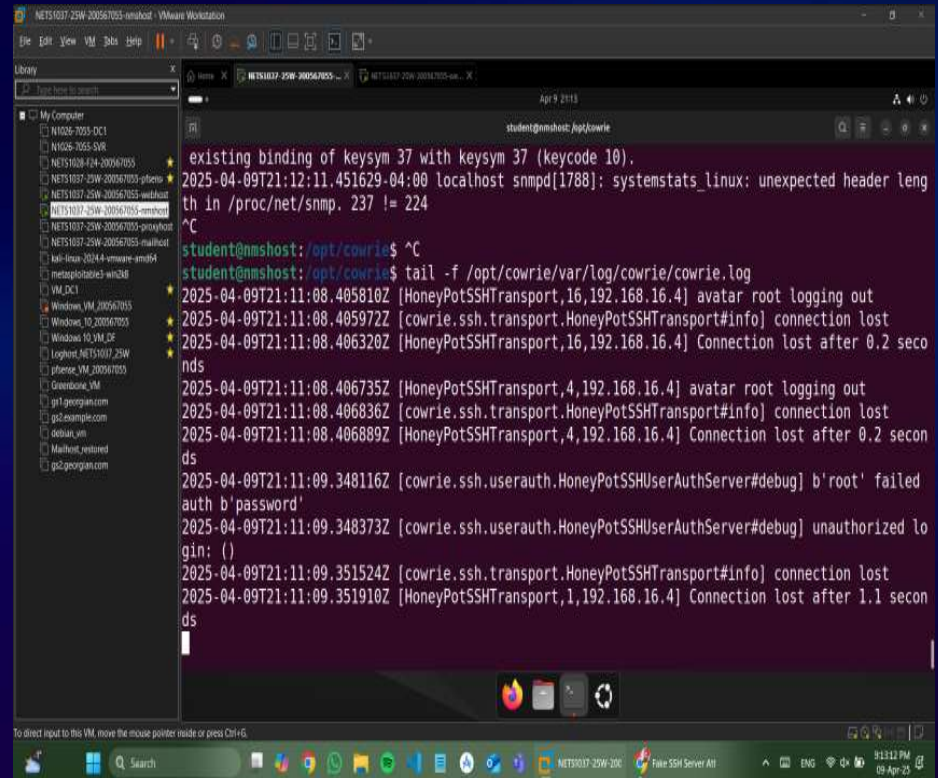
03

First point of attack identified

First point of Attack identified:

Detection of First Attack Attempt:

- Cowrie logs (``cowrie.log``) show first login attempt.
- Timestamp, IP, and attempted credentials captured.

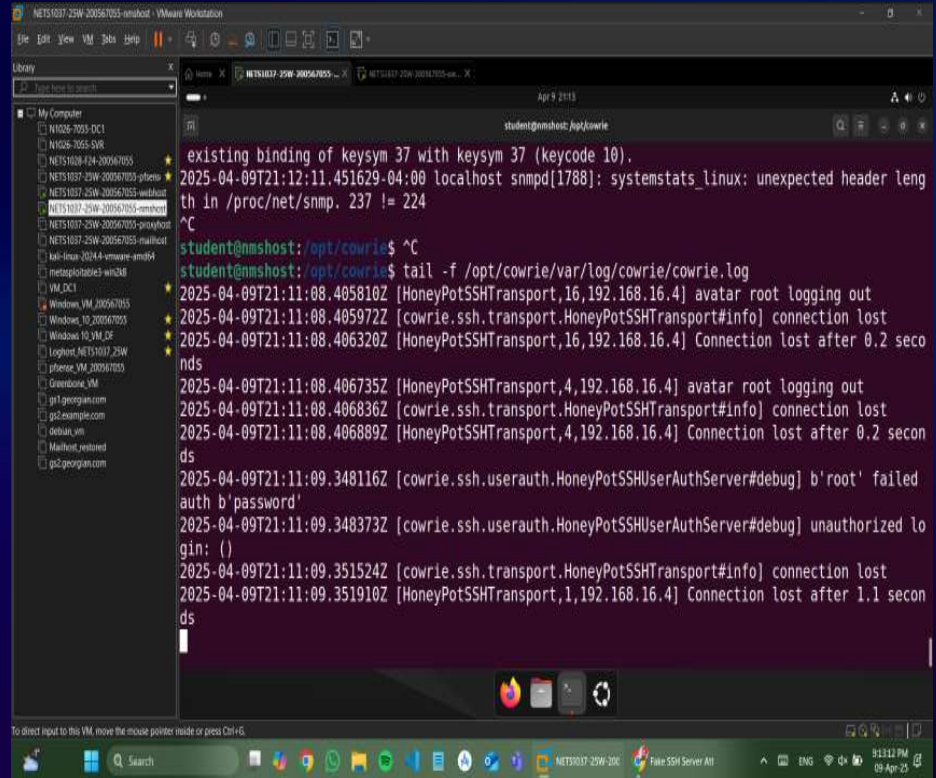


First point of Attack identified:

What it shows in this image is the credentials attacker is trying to use such as: user **“root”**, **“password”** in the attack.

While it failed, saying ***b 'root' failed auth b 'password' (line 7).***

It is the honeypot (fake server) that catches the honey 😊 ***“pardon my language”.***

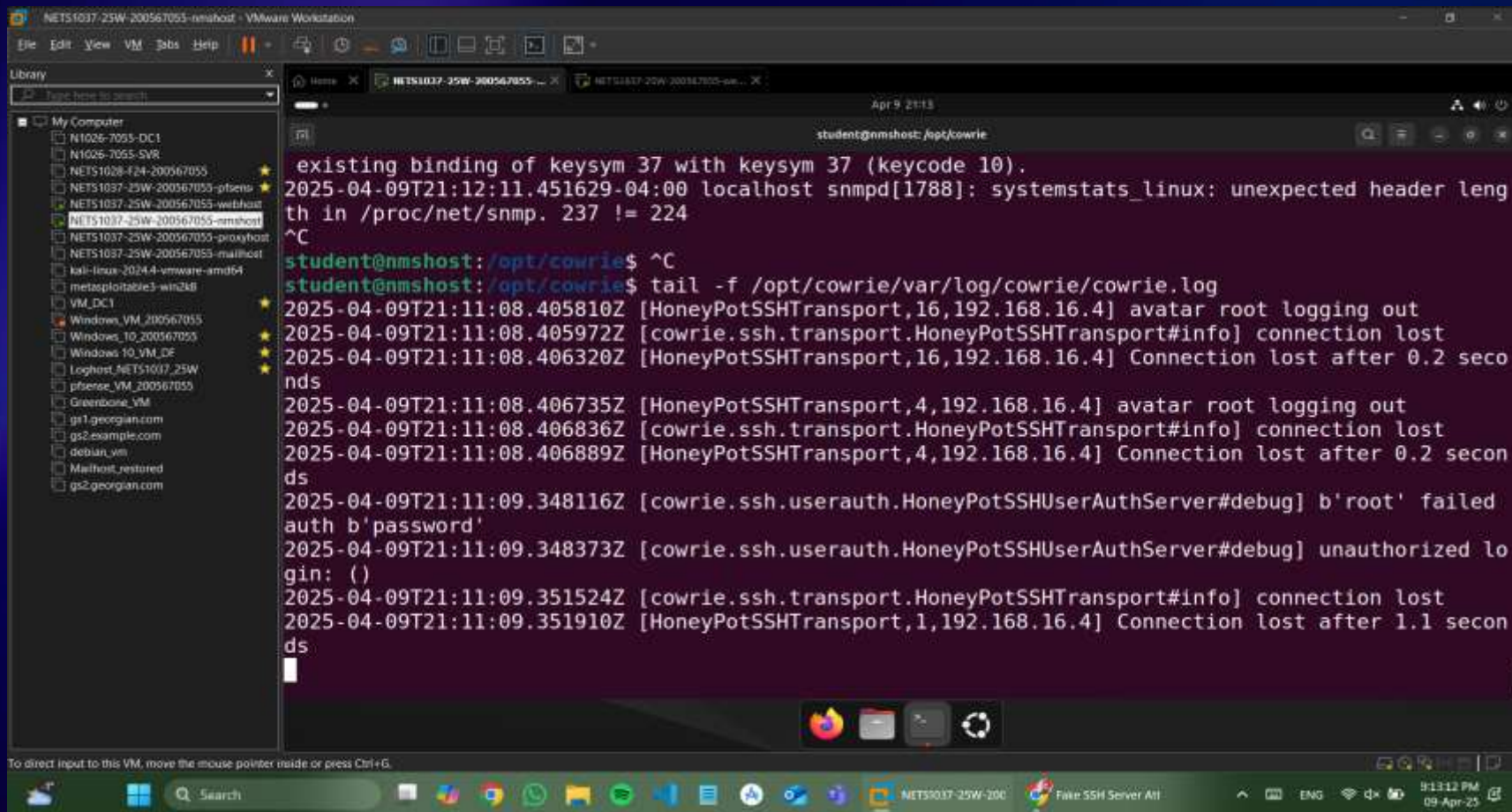


The screenshot shows a terminal window titled 'NETS1037-25W-200567055-remikost - VMware Workstation'. The terminal output shows a series of log messages and commands. The user 'student@nmshost: /opt/cowrie' is logged in. The terminal shows the following sequence of events:

- existing binding of keysym 37 with keysym 37 (keycode 10).
- 2025-04-09T21:12:11.451629-04:00 localhost snmpd[1788]: systemstats_linux: unexpected header length in /proc/net/snmp. 237 != 224
- ^C
- student@nmshost: /opt/cowrie\$ ^C
- student@nmshost: /opt/cowrie\$ tail -f /opt/cowrie/var/log/cowrie/cowrie.log
- 2025-04-09T21:11:08.405810Z [HoneyPotSSHTransport,16,192.168.16.4] avatar root logging out
- 2025-04-09T21:11:08.405972Z [cowrie.ssh.transport.HoneyPotSSHTransport#info] connection lost
- 2025-04-09T21:11:08.406320Z [HoneyPotSSHTransport,16,192.168.16.4] Connection lost after 0.2 seconds
- 2025-04-09T21:11:08.406735Z [HoneyPotSSHTransport,4,192.168.16.4] avatar root logging out
- 2025-04-09T21:11:08.406836Z [cowrie.ssh.transport.HoneyPotSSHTransport#info] connection lost
- 2025-04-09T21:11:08.406889Z [HoneyPotSSHTransport,4,192.168.16.4] Connection lost after 0.2 seconds
- 2025-04-09T21:11:09.348116Z [cowrie.ssh.userauth.HoneyPotSSHUserAuthService#debug] b'root' failed auth b'password'
- 2025-04-09T21:11:09.348373Z [cowrie.ssh.userauth.HoneyPotSSHUserAuthService#debug] unauthorized login: ()
- 2025-04-09T21:11:09.351524Z [cowrie.ssh.transport.HoneyPotSSHTransport#info] connection lost
- 2025-04-09T21:11:09.351910Z [HoneyPotSSHTransport,1,192.168.16.4] Connection lost after 1.1 seconds

The terminal window is running on a VMware Workstation. The left sidebar shows the library of virtual machines, including 'NETS1037-25W-200567055-remikost'. The bottom status bar shows the time as 11:12 PM on 09 Apr 25.

Monitored logs and captured attacker inputs:



The screenshot displays a VMware Workstation interface. On the left, a file explorer shows the contents of the 'My Computer' directory, listing various virtual machines and files. The main window shows a terminal session for the 'NETS1037-25W-200567055-nmshost' VM. The terminal output shows the following sequence of events:

```
existing binding of keysym 37 with keysym 37 (keycode 10).
2025-04-09T21:12:11.451629-04:00 localhost snmpd[1788]: systemstats_linux: unexpected header length in /proc/net/snmp. 237 != 224
^C
student@nmshost:/opt/cowrie$ ^C
student@nmshost:/opt/cowrie$ tail -f /opt/cowrie/var/log/cowrie/cowrie.log
2025-04-09T21:11:08.405810Z [HoneyPotSSHTransport,16,192.168.16.4] avatar root logging out
2025-04-09T21:11:08.405972Z [cowrie.ssh.transport.HoneyPotSSHTransport#info] connection lost
2025-04-09T21:11:08.406320Z [HoneyPotSSHTransport,16,192.168.16.4] Connection lost after 0.2 seconds
2025-04-09T21:11:08.406735Z [HoneyPotSSHTransport,4,192.168.16.4] avatar root logging out
2025-04-09T21:11:08.406836Z [cowrie.ssh.transport.HoneyPotSSHTransport#info] connection lost
2025-04-09T21:11:08.406889Z [HoneyPotSSHTransport,4,192.168.16.4] Connection lost after 0.2 seconds
2025-04-09T21:11:09.348116Z [cowrie.ssh.userauth.HoneyPotSSHUserAuthServer#debug] b'root' failed auth b'password'
2025-04-09T21:11:09.348373Z [cowrie.ssh.userauth.HoneyPotSSHUserAuthServer#debug] unauthorized login: ()
2025-04-09T21:11:09.351524Z [cowrie.ssh.transport.HoneyPotSSHTransport#info] connection lost
2025-04-09T21:11:09.351910Z [HoneyPotSSHTransport,1,192.168.16.4] Connection lost after 1.1 seconds
```

The terminal window also shows the user prompt 'student@nmshost:/opt/cowrie\$' and the command 'tail -f /opt/cowrie/var/log/cowrie/cowrie.log' being executed. The VMware interface includes a menu bar (File, Edit, View, VM, Tools, Help) and a status bar at the bottom with system icons and a clock showing 9:13:12 PM on 09-Apr-25.

04

Attack tracks in logs investigated

Tools used for logging:

- Logwatch
- Fwlog watch
- Cowrie's native logs
- Attack specific logs

Fact:

Why logging tools are important:

Logging tools such as **logwatch**, **fwlogwatch**, and **rsylogs** are essential as they enable organizations to track suspicious activity, detect attacks like brute-force attempts, and ensure compliance with security policies.

Analysis:

- Logwatch identified suspicious SSH activity.
- fwlogwatch identifies firewall related logs.
- cowrie native logs show all the attempts being done to the target machine.
- attack on the server (`nmshost`) collected and displayed all logs.

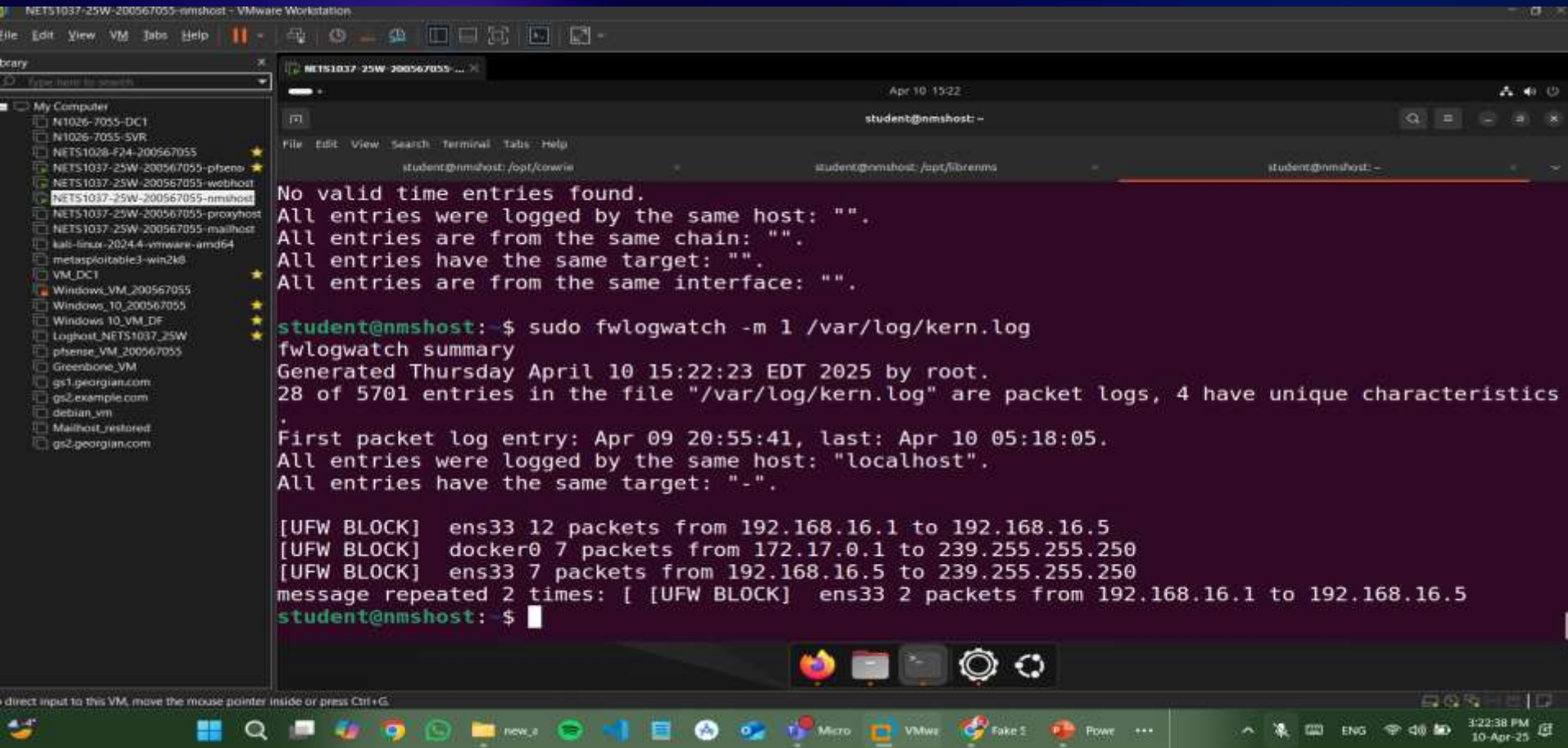
Logwatch-> suspicious activity:

The screenshot shows a VMware Workstation window titled "NETS1037-25W-200567055-nmshost - VMware Workstation". The interface includes a menu bar (File, Edit, View, VM, Tabs, Help), a toolbar, and a left-hand library pane. The library pane shows a tree view of the host's file system, with "My Computer" expanded. The main window displays a terminal window titled "student@nmshost: /opt/cowrie". The terminal output shows Logwatch results for the "sshd" service, indicating suspicious activity from 192.168.16.4.

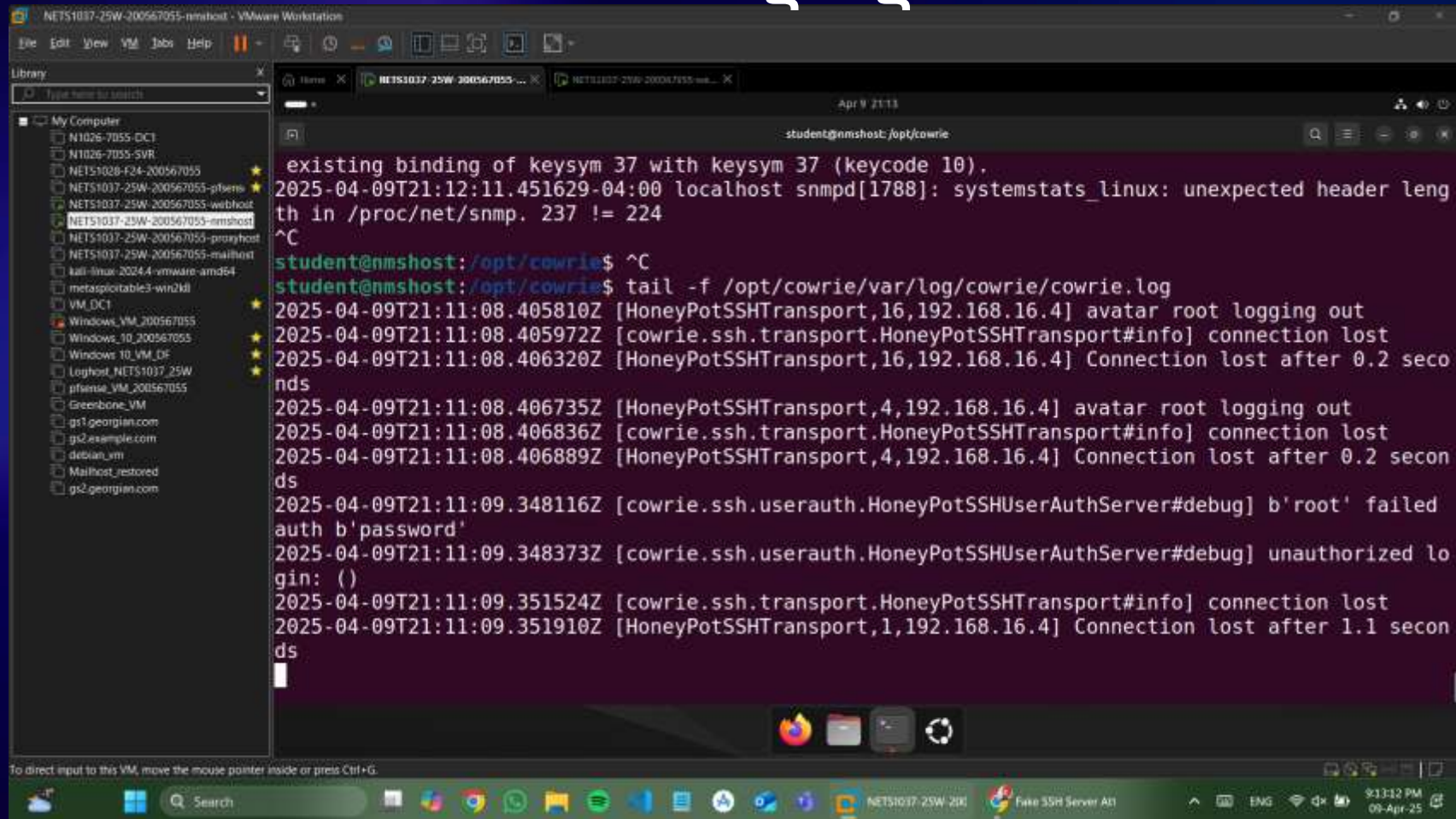
```
----- SSHD Begin -----  
  
SSHD Started: 1 Time  
  
Disconnecting after too many authentication failures for user:  
root : 14 Times  
  
Failed logins from:  
192.168.16.4: 116 Times  
root/password: 116 Times  
  
Received disconnect:  
Bye Bye [preauth]  
192.168.16.4 : 2 Time(s)  
  
**Unmatched Entries**  
drop connection #10 from [192.168.16.4]:45610 on [192.168.16.5]:22 past MaxStartups : 1 Time  
drop connection #11 from [192.168.16.4]:52190 on [192.168.16.5]:22 past MaxStartups : 1 Time  
error: beginning MaxStartups throttling : 2 Times  
exited MaxStartups throttling after 00:01:06, 2 connections dropped : 1 Time  
  
----- SSHD End -----
```

At the bottom of the terminal window, there is a status bar with icons for Firefox, a file manager, and a terminal. The system tray at the bottom of the screen shows the date and time as "1:52:48 AM".

Fwlogwatch showing logs:



Cowrie showing logs:



The screenshot shows a VMware Workstation interface with a terminal window titled "student@nmshost: /opt/cowrie". The terminal displays the following output:

```
existing binding of keysym 37 with keysym 37 (keycode 10).
2025-04-09T21:12:11.451629-04:00 localhost snmpd[1788]: systemstats_linux: unexpected header leng
th in /proc/net/snmp. 237 != 224
^C
student@nmshost:/opt/cowrie$ ^C
student@nmshost:/opt/cowrie$ tail -f /opt/cowrie/var/log/cowrie/cowrie.log
2025-04-09T21:11:08.405810Z [HoneyPotSSHTransport,16,192.168.16.4] avatar root logging out
2025-04-09T21:11:08.405972Z [cowrie.ssh.transport.HoneyPotSSHTransport#info] connection lost
2025-04-09T21:11:08.406320Z [HoneyPotSSHTransport,16,192.168.16.4] Connection lost after 0.2 seco
nds
2025-04-09T21:11:08.406735Z [HoneyPotSSHTransport,4,192.168.16.4] avatar root logging out
2025-04-09T21:11:08.406836Z [cowrie.ssh.transport.HoneyPotSSHTransport#info] connection lost
2025-04-09T21:11:08.406889Z [HoneyPotSSHTransport,4,192.168.16.4] Connection lost after 0.2 secon
ds
2025-04-09T21:11:09.348116Z [cowrie.ssh.userauth.HoneyPotSSHUserAuthServer#debug] b'root' failed
auth b'password'
2025-04-09T21:11:09.348373Z [cowrie.ssh.userauth.HoneyPotSSHUserAuthServer#debug] unauthorized lo
gin: ()
2025-04-09T21:11:09.351524Z [cowrie.ssh.transport.HoneyPotSSHTransport#info] connection lost
2025-04-09T21:11:09.351910Z [HoneyPotSSHTransport,1,192.168.16.4] Connection lost after 1.1 secon
ds
```

The VMware Workstation interface shows a library of virtual machines on the left, including "NETS1037-25W-200567055-nmshost". The bottom status bar indicates the VM is powered on and shows the system clock as 9:13:12 PM on 09-Apr-25.

/var/log/auth.log showing attack attempts:

```

command 'sudo' from deb sudo (1.9.14p2-lubuntul)
command 'sudo' from deb sudo-ldap (1.9.14p2-lubuntul)
Try: sudo apt install <deb name>
student@nmshost:~$ sudo grep "sshd" /var/log/auth.log
2025-04-09T21:16:20.209666-04:00 localhost sudo: student : TTY=pts/0 ; PWD=/opt/cowrie ; USER=root ; COMMAND=/usr/sbin/logwatch --detail High --range today --service sshd --logfile /opt/cowrie/var/log/cowrie.log
2025-04-10T01:31:12.009482-04:00 localhost sshd[4571]: Server listening on :: port 22.
2025-04-10T01:31:12.126277-04:00 localhost sshd[4573]: Received disconnect from 192.168.16.4 port 45524:11: Bye Bye [preauth]
2025-04-10T01:31:12.126357-04:00 localhost sshd[4573]: Disconnected from authenticating user root 192.168.16.4 port 45524 [preauth]
2025-04-10T01:31:12.457852-04:00 localhost sshd[4571]: error: beginning MaxStartups throttling
2025-04-10T01:31:12.457968-04:00 localhost sshd[4571]: drop connection #10 from [192.168.16.4]:45610 on [192.168.16.5]:22 past MaxStartups
2025-04-10T01:31:12.519387-04:00 localhost sshd[4583]: pam_unix(sshd:auth): authentication failure; logname= uid=0 euid=0 tty=ssh ruser= rhost=192.168.16.4 user=root
2025-04-10T01:31:12.520748-04:00 localhost sshd[4582]: pam_unix(sshd:auth): authentication failure; logname= uid=0 euid=0 tty=ssh ruser= rhost=192.168.16.4 user=root
2025-04-10T01:31:12.524337-04:00 localhost sshd[4577]: pam_unix(sshd:auth): authentication failure; logname= uid=0 euid=0 tty=ssh ruser= rhost=192.168.16.4 user=root

```

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

05

Monitoring tools checked for attack visibility

Ntopng and LibreNMS

are both network monitoring tools, but they serve different purposes and provide different features.

Ofcourse, they are very good tools to use for monitoring purposes.

I am using them for capturing network flow(in case of ntopng) ,and showing traffic logs (in case of librenms) for my attack.

Ntopng showing live flows (SSH):

NETS1037-ZSW-200567055-nmshost - VMware Workstation

File Edit View VM Tools Help

Library

- My Computer
 - N1026-7055-DC1
 - N1026-7055-SVR
 - NETS1028-F24-200567055
 - NETS1037-ZSW-200567055-pfsens
 - NETS1037-ZSW-200567055-webhost
 - NETS1037-ZSW-200567055-nmshost
 - NETS1037-ZSW-200567055-proxyhost
 - NETS1037-ZSW-200567055-mailhost
 - kali-linux-20244-vmware-amd64
 - metasploitable3-win2k8
 - VM_DC1
 - Windows_VM_200567055
 - Windows_10_200567055
 - Windows_10_VM_DF
 - Loghost_NETS1037_ZSW
 - pfsense_VM_200567055
 - Greenbone_VM
 - gs1.georgian.com
 - gs2.example.com
 - debian_vm
 - Mailhost_restored
 - gs2.georgian.com

Dashboard Monitoring Alerts Hosts Maps Interface Settings Developer Help

Problem loading page: x ntopng - Live

Apr 10 04:39

Not Secure 192.168.16.4:3000/lua/flows_stats.lua?alert_type=warning&flowhosts_type=AllGroups&Application=Alert_type

ens33 19.00 Kbps 6.70 Kbps Upgrade to Pro/Enterprise version

Live Flows | Analysis

Host	Protocol	Application	Status	TCP Flow State	Traffic Type	Host Pools	Networks	Result
All	All	All	All	All	All	All	All	Result

Actions	Last Seen	Duration	Protocol	Score	Flow	Actual Thpt	Total By...	Info
	00:11	00:11	TCP/HTTP:ntop...	60	nmshost [L] : 59690 ↔ webhost [L] : ...		65.48 KB	GET OK 192.168.16.4:3000/lua/flows_sta
	00:30	00:30	TCP/HTTP:ntop...	60	nmshost [L] : 33626 ↔ webhost [L] : ...		12.70 KB	GET OK 192.168.16.4:3000/lua/rest/v2/g
	00:20	00:20	TCP/HTTP:ntop...	60	nmshost [L] : 58736 ↔ webhost [L] : ...		12.69 KB	GET OK 192.168.16.4:3000/lua/rest/v2/g
	00:10	00:10	TCP/HTTP:ntop...	60	nmshost [L] : 59730 ↔ webhost [L] : ...		11.93 KB	GET OK 192.168.16.4:3000/lua/rest/v2/g
	00:10	00:11	TCP/HTTP:ntop...	60	nmshost [L] : 59706 ↔ webhost [L] : ...	17.61 Kbps ↑	7.25 KB	GET OK 192.168.16.4:3000/lua/rest/v2/g
	00:44	00:44	TCP/SSH	50	webhost [L] : 51496 ↔ nmshost [L] : ...		4.15 KB	
	00:44	00:44	TCP/SSH	50	webhost [L] : 51582 ↔ nmshost [L] : ...		4.15 KB	
	00:44	00:44	TCP/SSH	50	webhost [L] : 51546 ↔ nmshost [L] : ...		4.15 KB	
	00:44	00:44	TCP/SSH	50	webhost [L] : 51542 ↔ nmshost [L] : ...		4.15 KB	
	00:44	00:44	TCP/SSH	50	webhost [L] : 51528 ↔ nmshost [L] : ...		4.15 KB	

Showing page 1 of 4: total 31 rows

1 2 3 4

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

NETS1037- ntopng SSH ChatGPT- PowerB

4:39:26 AM 10-Apr-25

Webhost SSH attempts to Nmshost:

NETS1037-25W-200567055-nmshost - VMware Workstation

File Edit View VM Tabs Help

Library

- My Computer
 - N1026-7055-DC1
 - N1026-7055-SVR
 - NETS1028-F24-200567055
 - NETS1037-25W-200567055-pfsens
 - NETS1037-25W-200567055-webhost
 - NETS1037-25W-200567055-nmshost
 - NETS1037-25W-200567055-proxyhost
 - NETS1037-25W-200567055-mailhost
 - kali-linux-2024.4-vmware-arm64
 - metasploitable3-win2k8
 - VM_DC1
 - Windows_VM_200567055
 - Windows_10_200567055
 - Windows_10_VM_DF
 - Loghost_NETS1037_25W
 - pfsense_VM_200567055
 - Greenbone_VM
 - gs1.georgian.com
 - gs2.example.com
 - debian_vm
 - Mailhost_restored
 - gs2.georgian.com

Apr 10 15:43

Download Archives - Ads x Problem loading page x nitapng - Live x nitapng - Flow Details x +

192.168.16.4 3000/444flow_detail3anflow_key=21004007144flow_hash_id=31174580

ens33 27.60 Kbps 38.10 Kbps Upgrade to Pro/Enterprise version

This flow has been purged from memory.

Flow: 192.168.16.4:51116 ↔ 192.168.16.5:2222 | Overview

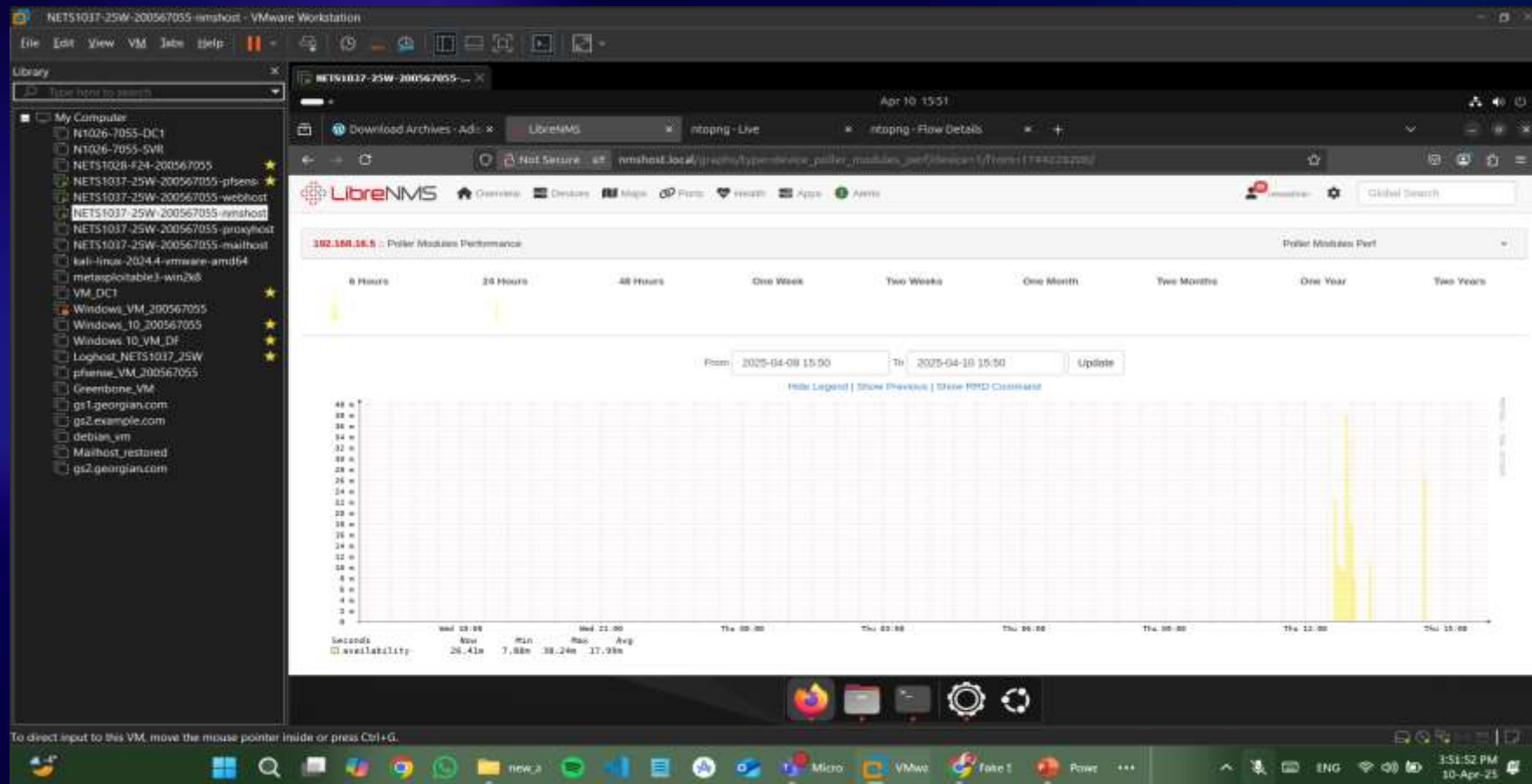
Flow Peers [Client / Server]	webhost [192.168.16.4:51116 [00:0C:29:9B:CC:C1] ↔ nmshost [192.168.16.5:2222 [00:0C:29:B9:C7:BF]
Protocol / Application	TCP / SSH (RemoteAccess) [Confidence: 100%]
First / Last Seen	10/04/2025 15:42:45 [00:58 sec ago] 10/04/2025 15:42:45 [00:58 sec ago]
Flow Duration	< 1 sec
Total Traffic	Total: 4.2 KB — Goodput: 2.4 KB (57.14%) — Client → Server: 12 Pkts / 2 KB — Client ← Server: 15 Pkts / 2.2 KB —
DSCP / ECN [Client / Server]	Best Effort [CS0] / Disabled (0) Best Effort [CS0] / Disabled (0)
RTT Time Breakdown	0.588 ms (persec)
Application Latency	57.19 ms
Packet Inter-Arrival Time [Min / Avg / Max]	Client → Server: < 1 ms / 34.6 ms / 103 ms Client ← Server: < 1 ms / 35.42 ms / 102 ms

To direct input to this VM, move the mouse pointer inside or press Ctrl+G

new.s Micro VMware View Flow

3:43:43 PM 10-Apr-25

Librenms showing some traffic here in the graphs:



Company name

06

Attack success/failure
identified

Why the Attack Failed for the Attacker:

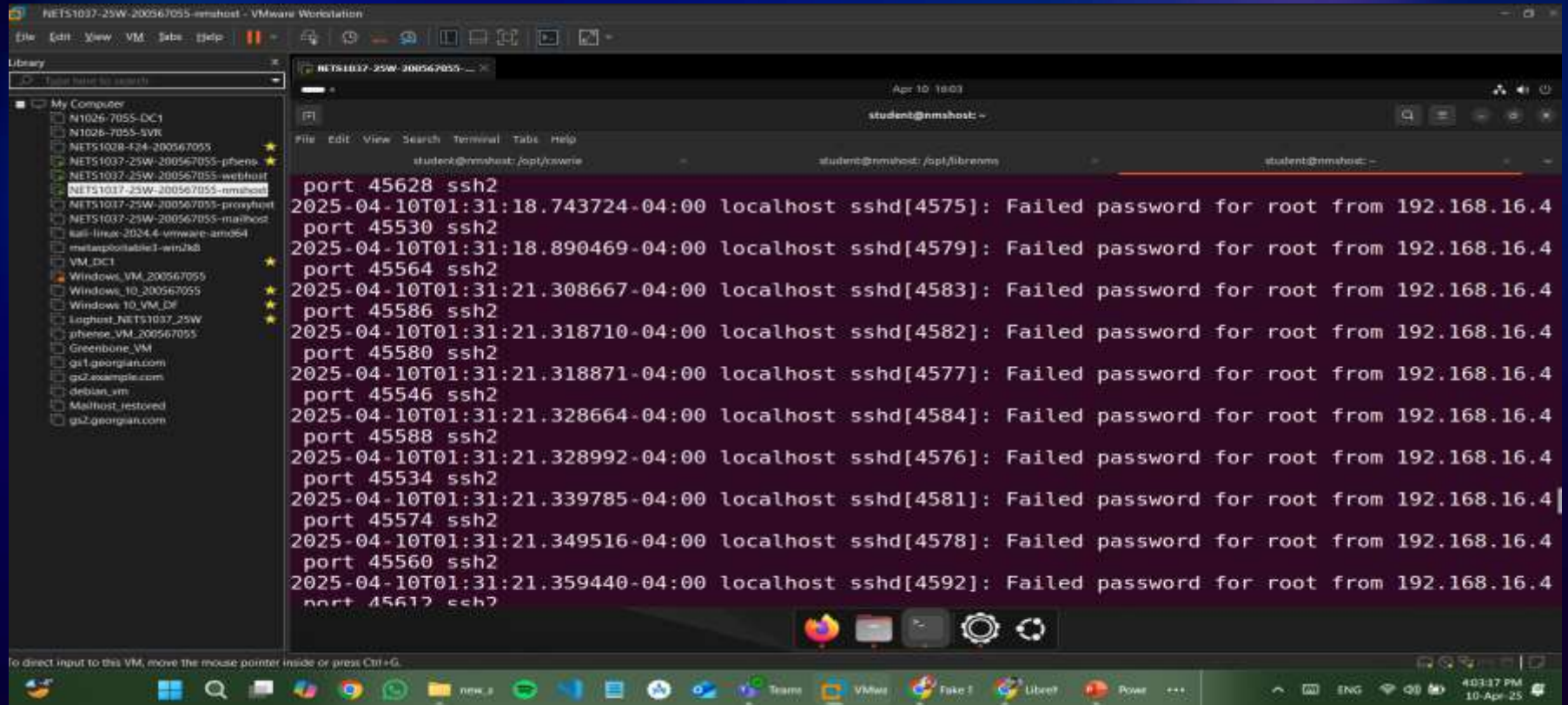
Cowrie Honeypot Detection: The attack was logged with full details (username, IP address, and failed password attempt) and effectively identified as a failed brute force attempt.

Incorrect Credentials: The attacker used invalid login credentials, which were rejected by both the Cowrie honeypot and the SSH service, ensuring no successful login was made.

When an attack fails, it's crucial to document the evidence and mechanisms that caused the failure, as well as steps that can be taken to improve security and prevent future failures from being successful

Evidence for the failure of the attack

Auth logs showing failed login attempt from another IP address



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Mitigation and Prevention Strategies

Mitigation and Prevention Strategies:

Strengthen Firewall (UFW) Settings:

- Limit SSH access to trusted IPs only, effectively reducing the attack surface.
- Example:

```
student@nmshost: sudo ufw allow from 192.168.1.0/24 to any port 22  
sudo ufw deny 2222
```

Implement Fail2Ban:

- Automatically block IP addresses after multiple failed login attempts, preventing brute force attacks.
- For that need to install Fail2Ban and do the configuration for banning repeated offenders.

Mitigation and Prevention Strategies:

Enhance Cowrie Honeypot Logging:

- Increase the detection and logging thresholds to alert administrators of potential threats faster.

Enable Multi-Factor Authentication (MFA) for SSH:

- Even if an attacker gains valid credentials, MFA provides an extra layer of protection.

Harden SSH Configuration:

- Disable password-based logins and enforce key-based authentication to further secure the system.
 - Example:

```
sudo nano /etc/ssh/sshd_config PasswordAuthentication no
```

Thanks

Any Questions?